AISHVARY TURKAR

M.Sc. MECHATRONICS | ASPIRING ROBOTICS SOFTWARE ENGINEER

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PROFILE

Mechatronics M.Sc. student at the University of Siegen with a Mechanical Engineering background and hands-on experience in manufacturing, automation, and SAP EWM. Commissioned CNC equipment, supported process validation and shop-floor improvement, and built robotics and embedded systems projects. Interested in internships in manufacturing, automation, and robotics.

SKILLS

Robotics & automation: robot programming, sensor integration, actuators, Arduino, ESP8266, PLC, inverse kinematics
Manufacturing & design: CNC commissioning, process optimization, CAD/CAM, GD&T, AutoCAD, 3D printing, SAP EWM
Programming & software: Python, C++, JavaScript, G-code, HTML, CSS, SCSS, WordPress, Arduino IDE, VS Code

EXPERIENCE

Assistant Manager - Manufacturing Operations

VE Commercial Vehicles Ltd. | Jun 2024 - Sep 2025

- Commissioned 8 CNC machines (6 HMC and 2 VMC), contributing to annual production capacity growth from 75,000 to 80,000 units (+6.7%).
- Managed installation, inspection, engineering trials, machine testing, and process validation for new equipment and manufacturing processes.
- Prepared 2D/3D drawings, layouts, fixtures, and process sheets in AutoCAD; conducted monthly process, tooling, and fixture audits.
- Supported standard operating procedures and continuous improvement, including a taper roller curve conveyor project that improved OLE by 4%.
- Led SAP EWM storage-system implementation for EURO-7 and CNG variants, improving inventory management and material flow.

Functional Trainee

VE Commercial Vehicles Ltd. | Jun 2023 - Jun 2024

- Completed a year of training before moving into manufacturing operations responsibilities.

Intern

Ministry of Cooperation | Dec 2022 - Jan 2023

- Assisted with district-level development of the National Cooperation Database and structured data management.

Web Designing Intern

Eurecon Pumps India Pvt. Ltd. | May 2022 - Jul 2022

- Supported website content development, branding, and digital design assets.

EDUCATION

M.Sc. Mechatronics - University of Siegen | Oct 2025 - Present
B.Tech. Mechanical Engineering - Samrat Ashok Technological Institute, Vidisha | 2019 - 2023 | Honours, final grade 1.6
Diploma in Computer Applications - Makhanlal Chaturvedi National University of Journalism and Communication | 2018 - 2019

PROJECTS & DEVELOPMENT

Selected engineering, robotics, and automation work

PROJECTS

Color-Based Object Sorting Manipulator

Robotics | LEGO EV3

- Developed a Python-controlled conveyor sorting manipulator and implemented inverse kinematics for pick-and-place movement.
- Integrated color, touch, and encoder sensors for perception, homing, and motion control; tested across different operating conditions.

5-DOF Robotic Arm

Robotics | Kinematics

- Designed and programmed a five-degree-of-freedom arm for pick-and-place automation.
- Implemented forward and inverse kinematics and developed an Android app for wireless operation.

IoT-Based Home Automation

Embedded systems | ESP8266

- Enabled voice-controlled remote operation of two LED devices using Google Assistant, Wi-Fi communication, and a two-channel relay module.

Digital Tachometer

Embedded systems | Arduino Nano

- Developed a PCB-based tachometer with an IR sensor and LCD to display live rotational-speed measurements.

CERTIFICATIONS

Problem Solving through Programming in C - NPTEL IIT Kharagpur
Computer Aided Designing on CATIA-V5 - S.A.T.I. Vidisha
Learning Arduino: Foundations - LinkedIn Learning
Fundamentals of Digital Marketing - Google Digital Garage
Learning Industrial Automation; PLC Ladder Logic 1 and 2 - LinkedIn Learning (2026)

LANGUAGES

English (C2 / full professional proficiency) | Hindi (native) | German (A1, currently improving)

ADDITIONAL INFORMATION

Interests: industrial robotics, automation, manufacturing systems, photography, chess, and DIY projects.